

Homework Submission Policy

- **Late submission:**
 - within 24 hours after its due will incur 20% penalty,
 - after 24 hours and within seven days of its due will incur 50% penalty, and
 - after seven days of its due will not be graded.

Note: One minute late is the same as 23 hours late.

Homework Assignment #4

Textbook Exercises (Based on the 9th edition textbook)

3.2-17. If 15 observations are taken ...

3.2-18. Say the serum cholesterol level ...

3.3-11. A candy maker produces mints ...

4.1-9. A manufactured item is classified ...

4.2-3. Roll a fair four-sided die ...

Due: 2:20pm, May 7, 2026

3.2-17. If 15 observations are taken independently from a chi-square distribution with 4 degrees of freedom, find the probability that at most 3 of the 15 observations exceed 7.779.

3.2-18. Say the serum cholesterol level (X) of U.S. males ages 25–34 follows a translated gamma distribution with pdf

$$f(x) = \frac{x - 80}{50^2} e^{-(x - 80)/50}, \quad 80 < x < \infty.$$

- (a) What are the mean and the variance of this distribution?
- (b) What is the mode?
- (c) What percentage have a serum cholesterol level less than 200? HINT: Integrate by parts.

3.3-11. A candy maker produces mints that have a label weight of 20.4 grams. Assume that the distribution of the weights of these mints is $N(21.37, 0.16)$.

- (a)** Let X denote the weight of a single mint selected at random from the production line. Find $P(X > 22.07)$.
- (b)** Suppose that 15 mints are selected independently and weighed. Let Y equal the number of these mints that weigh less than 20.857 grams. Find $P(Y \leq 2)$.

4.1-9. A manufactured item is classified as good, a “second,” or defective with probabilities $6/10$, $3/10$, and $1/10$, respectively. Fifteen such items are selected at random from the production line. Let X denote the number of good items, Y the number of seconds, and $15 - X - Y$ the number of defective items.

- (a)** Give the joint pmf of X and Y , $f(x, y)$.
- (b)** Sketch the set of integers (x, y) for which $f(x, y) > 0$.
From the shape of this region, can X and Y be independent? Why or why not?
- (c)** Find $P(X = 10, Y = 4)$.
- (d)** Give the marginal pmf of X .
- (e)** Find $P(X \leq 11)$.

4.2-3. Roll a fair four-sided die twice. Let X equal the outcome on the first roll, and let Y equal the sum of the two rolls.

- (a)** Determine μ_X , μ_Y , σ_X^2 , σ_Y^2 , $\text{Cov}(X, Y)$, and ρ .
- (b)** Find the equation of the least squares regression line and draw it on your graph. Does the line make sense to you intuitively?